

A two digit BCD Downcounter

We would like to design a 2-digit BCD downcounter that counts down from 59. You are provided with a 4-digit binary downcounter with parallel load and count enable in *downcounter.dig*. The state transition table of this counter is given in Table 1.

LOAD	EN	CNT(t+1)
1	X	IN
0	0	CNT(t)
0	1	CNT(t)-1

Table 1: Downcounter with parallel load and enable.

Pre-lab assignment

The file *down59.dig* contains two copies of *downcounter.dig*, which are simply interconnected to demonstrate their operation. A test bench has been created in *down59test.dig*. In its current setting, the counter generates the sequence FF, EE, DD,... Fig. 1 shows the counter and the test bench.

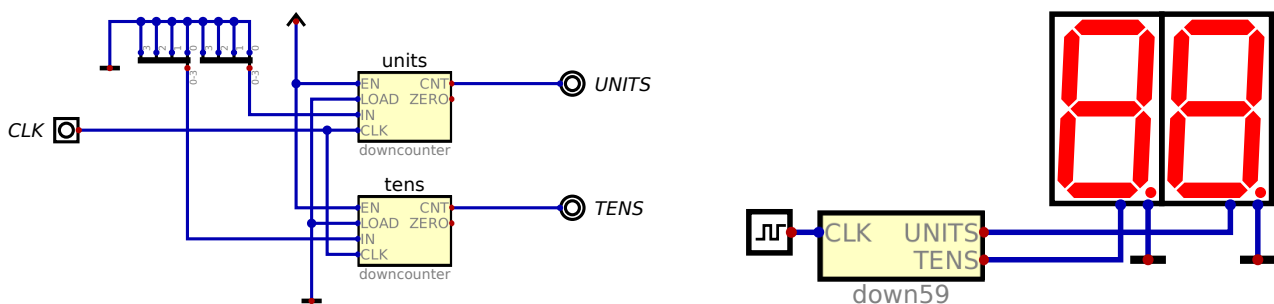


Fig. 1: Two digit downcounter (left), test bench for the circuit to be designed (right).

Modify *down59.dig* so that it counts the BCD sequence 59, 58, 57, ..., 01, 00, 59, ... Upload the modified version as your preliminary work.

In-lab

During the laboratory session, you will test your circuit on the Nexys A7 board. You will be then asked to make a modification to the circuit to implement a different function. You may work in groups of two. Bring along your computer with all the software tools installed.

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